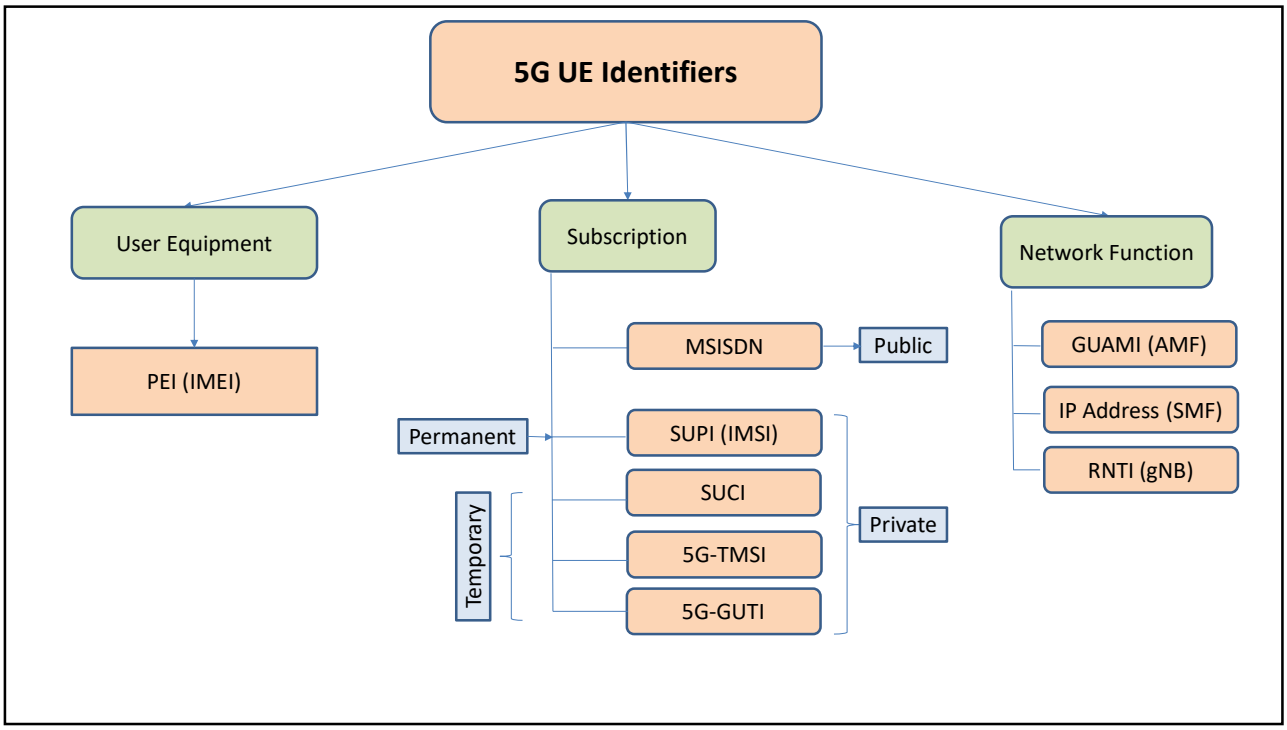
5G Core Network Procedures

Agenda

- 5G Core Procedures
  - UE Registration – Authentication, NAS security, AN context setup
  - Mobility Management Procedures: Mobility Registration
  - AN Link management Procedures – AN release, Service request, Paging
- Data Path Handling Procedures
  - Session Management Procedure: Session Setup/Modification/Release
  - QoS flow – creation/Modification/deletion



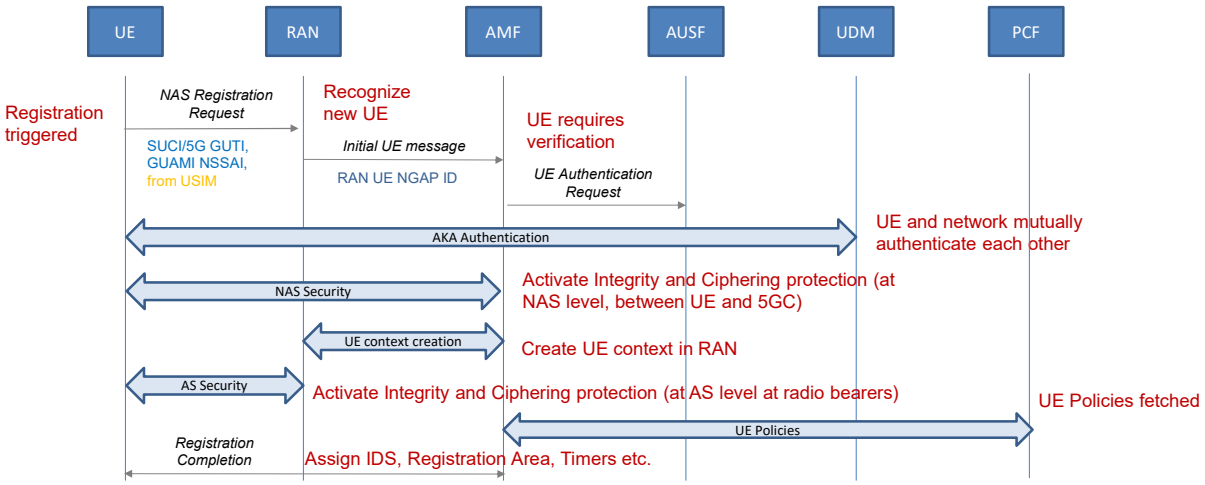
UE Registration & De-Registration – Why

- Used to register or deregister a UE/user within the 5G System
- UE provides the network with its identity and information e.g SUPI/SUCI, 5G-GUTI
- Authenticates itself and the Network – 5G-AKA,
- Requests for Services, Slices and gets access to authorized ones - NSSAI
- Gets reachability factors to stay informed to network about it's where-abouts - TAI
- Periodically establishes the user context within the 5G System
- Network De-registers UE if not traceable within the system for a duration
  - Re-Registration by UE essential to gain access again
- UE requests for De-registration, at UE power-off and when no more 5G services are needed. E.g. Aeroplan mode

UE Registration – When

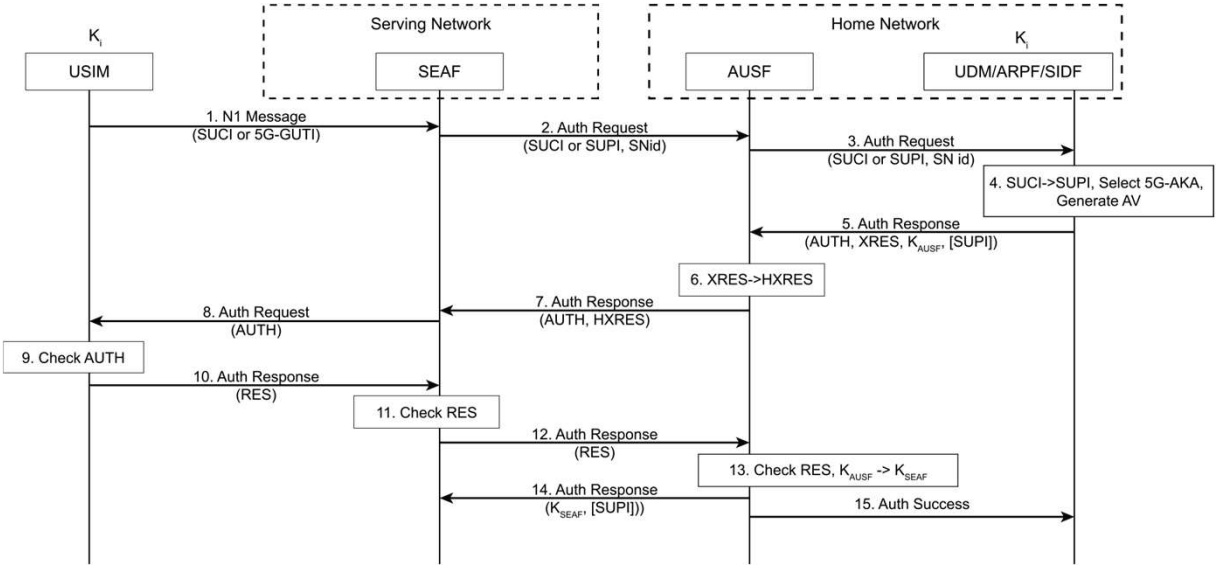
- UE initiates registration procedures in the following scenarios:
  - **Initial registration**: At power up or first contact with the network, to create the new UE context within the network
  - **Mobility registration**: To enable tracking and ensure reachability. Done on moving to a new Tracking Area
  - **Periodic Registration Update**: Done after a predefined time of inactivity
  - To update its **capabilities** or protocol parameters that are negotiated in Registration procedure with or without changing to a new TA
  - **Emergency Registration**: Registration by UEs requiring to perform emergency services when it cannot gain normal services from the network

UE Initial Registration - Overview

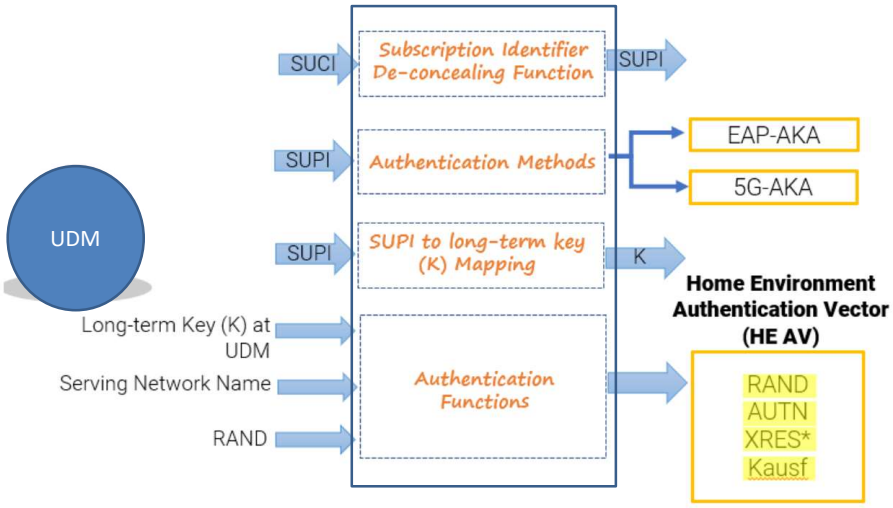


Informations Provided by 5GC to UE: 5G GUTI (GUAMI + 5G TMSI), NSSAIs, TAI list, UE Policies, Equivalent PLMNs

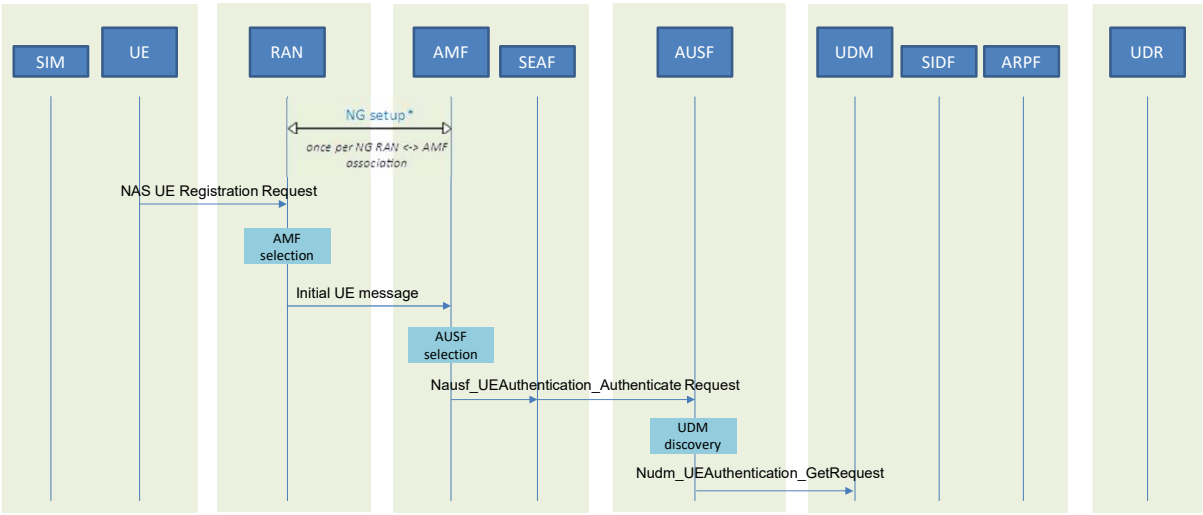
UE Authentication procedure



UE Initial Registration – Authentication Vector Summary



UE Initial Registration – Details



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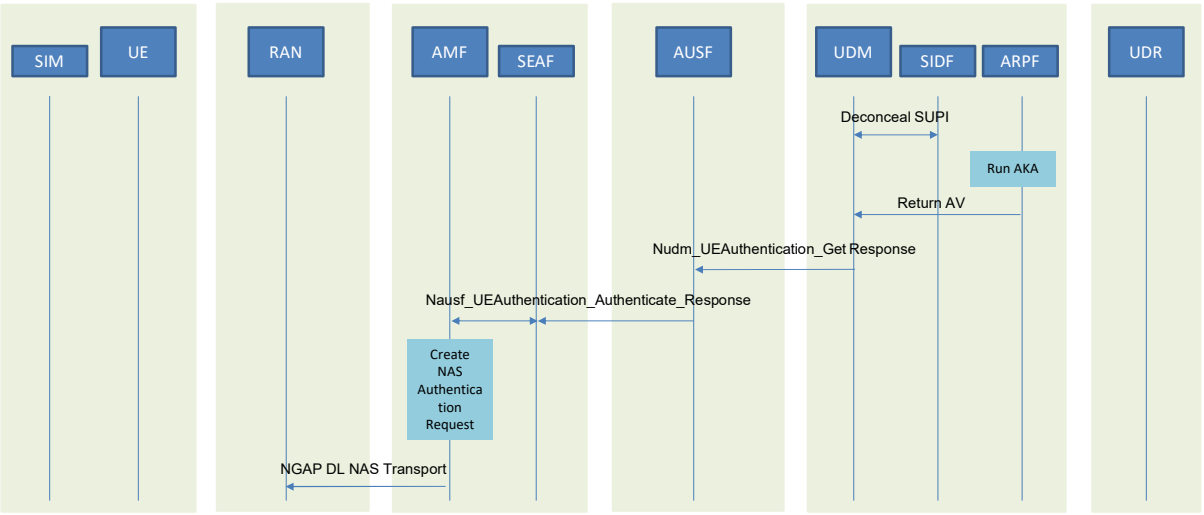

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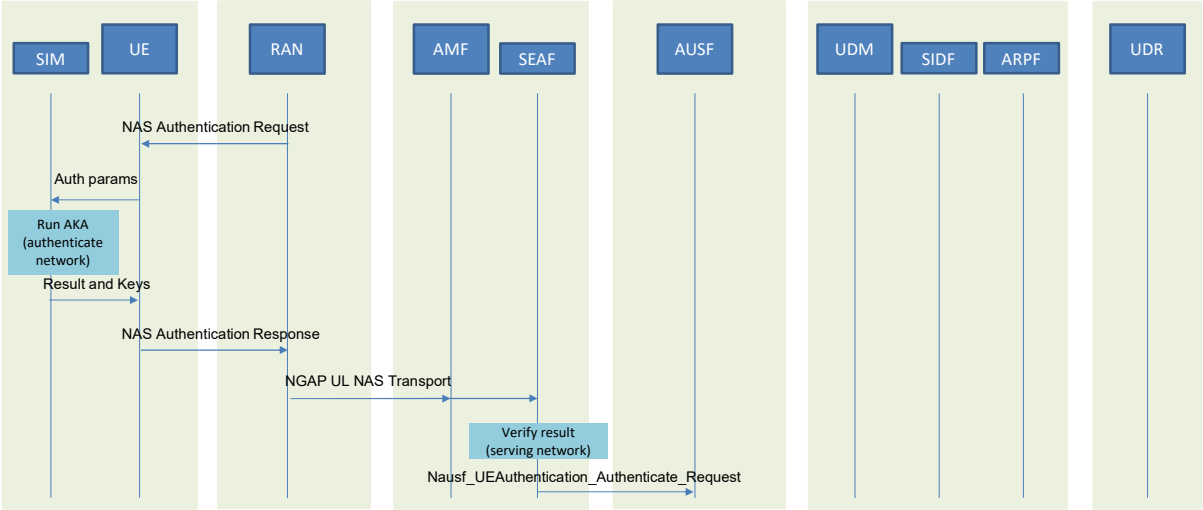

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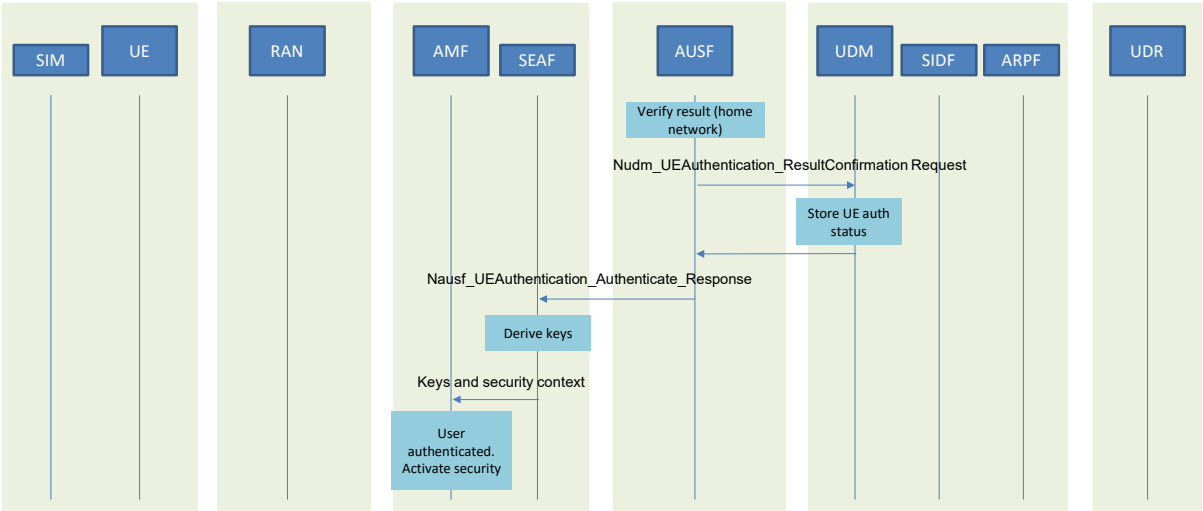

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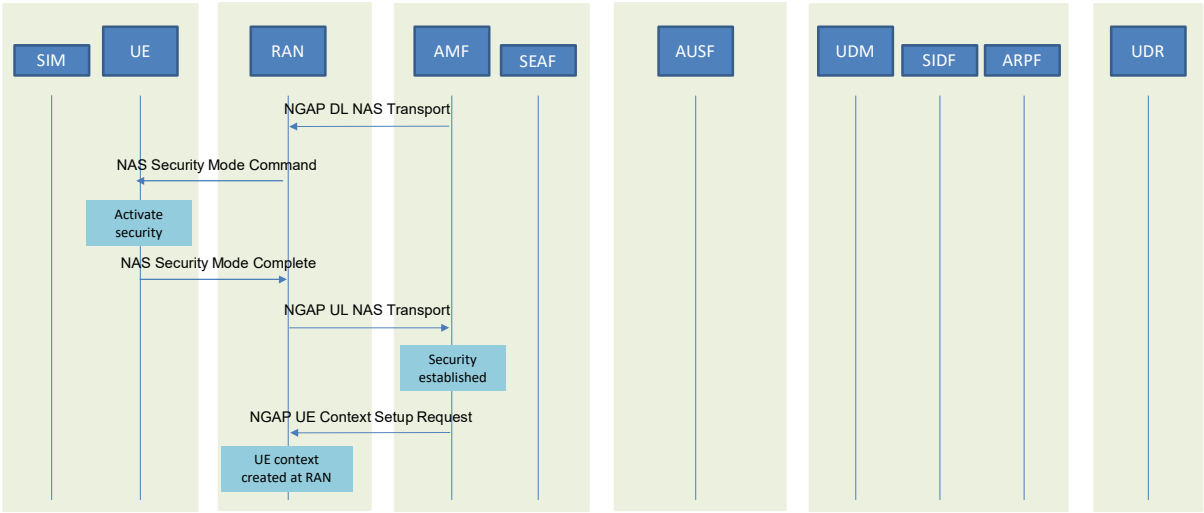

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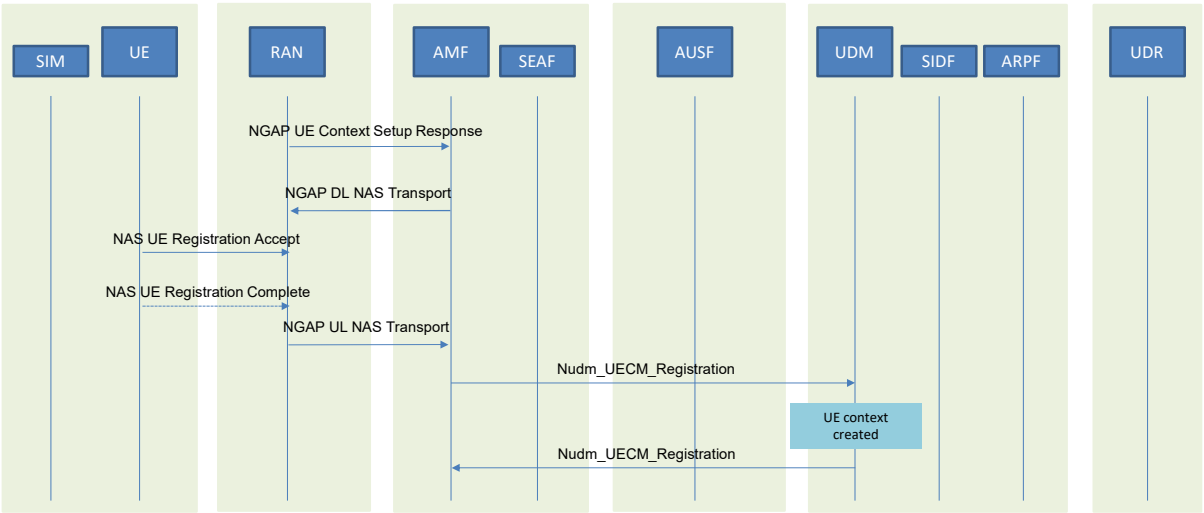
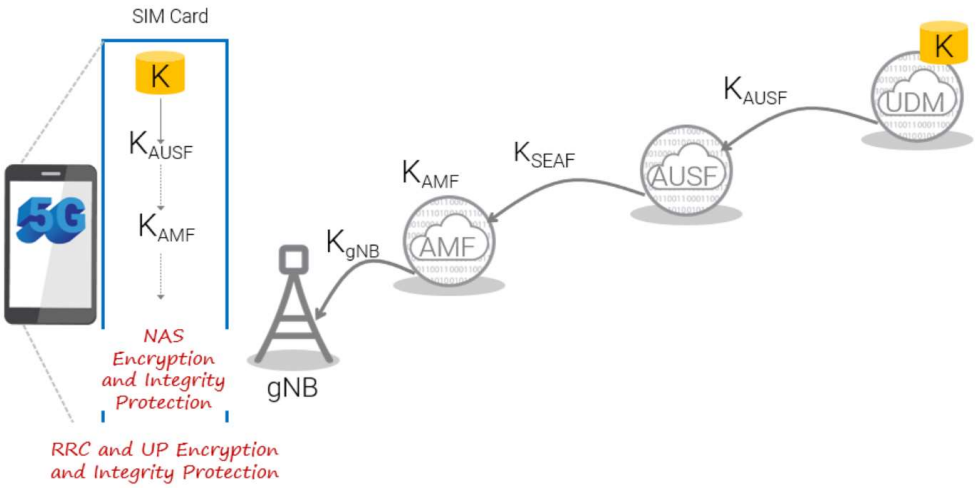


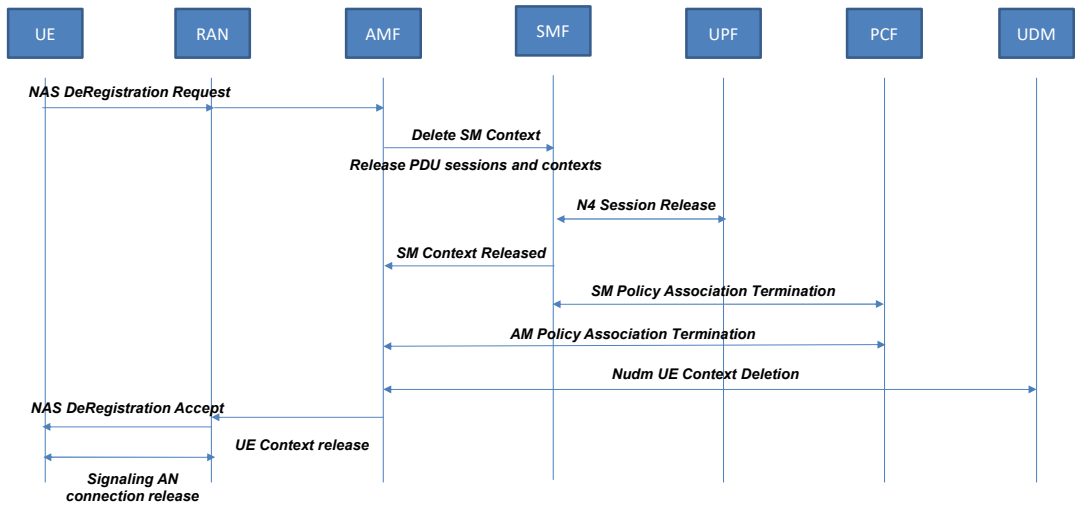
UE Initial Registration – Security Keys



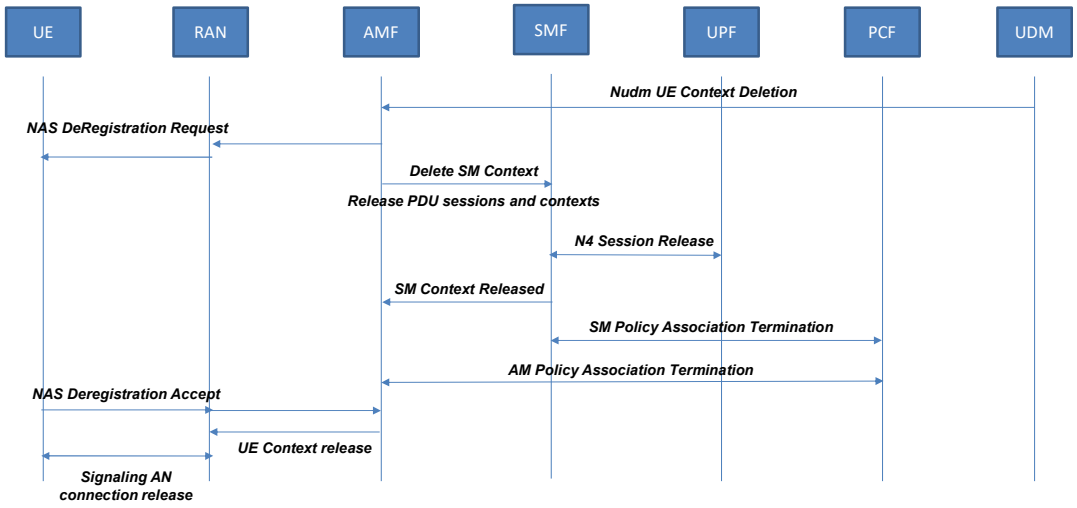
UE De-registration

- Deregistration procedure allows:
- UE to inform the network that it does not want to access the 5GS any longer,
  - e.g. on switch off, airplane mode on, SIM removal
- Network to inform the UE that it does not have access to the 5GS any longer
  - e.g. Subscription change, UE not reachable

UE Initiated De-Registration



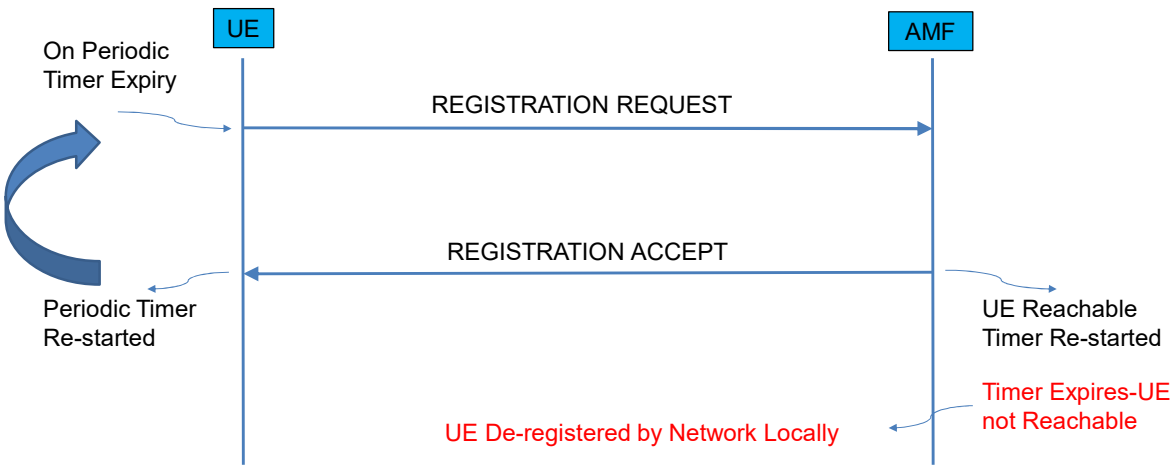
Network Initiated De-Registration



Other UE Registration Procedures

- Periodic Registration:
  - WHY?
    - UE IDLE state “I am alive” & “I am here” indication to Network
    - UE over 3GPP access periodically registers to the Core to update its whereabouts
    - Not applicable to non 3gpp access
  - WHEN?
    - Periodic registration timer provided by AMF during Initial Registration
    - Timer started when UE enters CM-IDLE state in RM-REGISTERED state
    - Timer Stopped when UE enters CM-CONNECTED / RM-DEREGISTERED state
  - HOW?
    - Periodic registration performed by UE upon periodic timer expiry
    - On successful completion, timer restarted in UE with the initial timer value
    - Similarly “UE Mobile Reachable timer” run by the AMF, re-started when UE communicates.
      - UE De-registered by Network if that timer expires
        - UE is not reachable

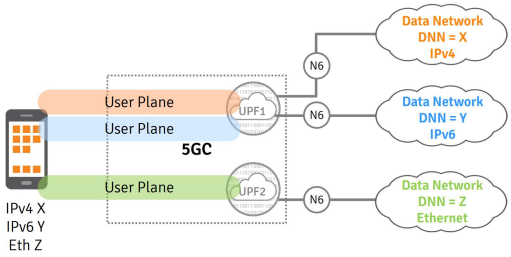
UE Period Registration

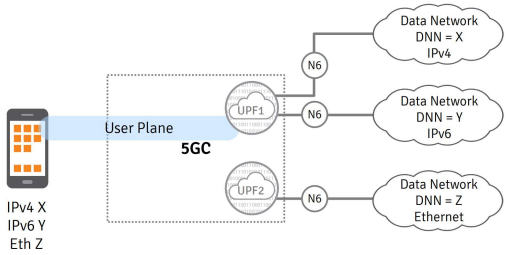


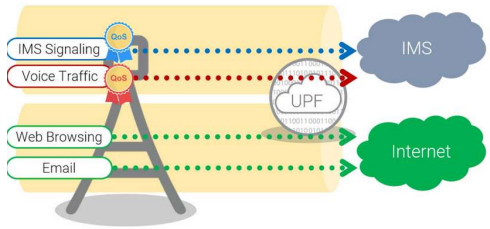
Other UE Registration Procedures

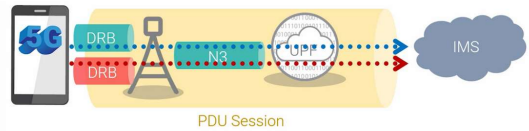
- Mobility Registration:
  - WHY?
    - UE to Inform Network on it's Location change – "I am out of RA"
    - To facilitate effective Paging
  - WHEN?
    - Tracking Area list (RA) provided by Network, within which UE is free to roam without Registering
      - Provided during Initial Registration to UE
    - One entering new TA, UE performs Mobility registration and gets Area specific informations
      - Dynamically calculated based on various factors and updated during every Mobility registration

PDU Session Procedures

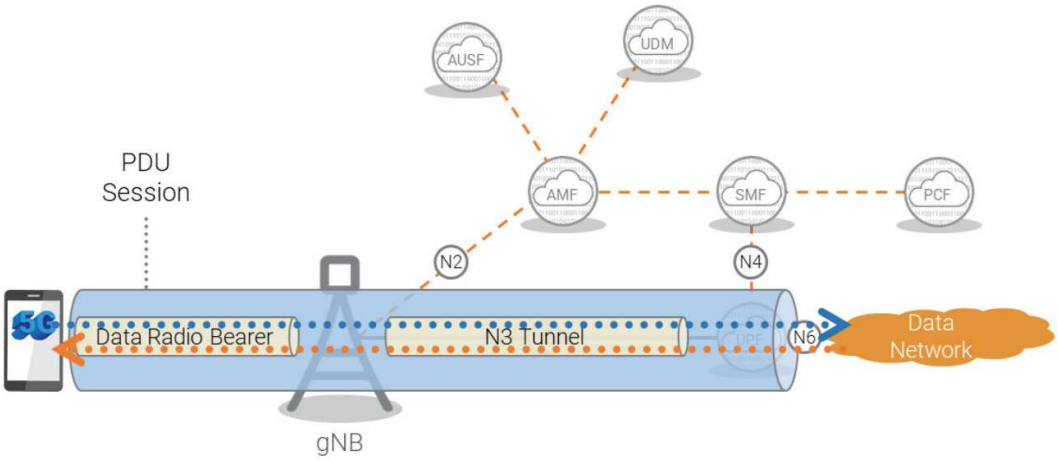








PDU Session Procedures

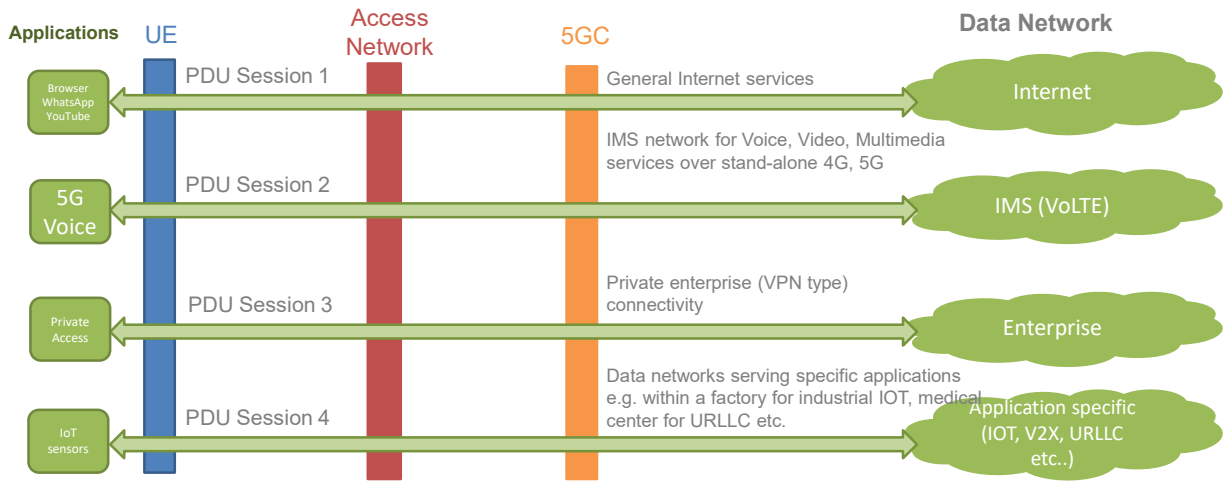


PDU Session - Introduction

- PDU Session: Connection from the UE to a Data Network Name(DNN) via one or more access networks
- Data Network: a network that provides access to specific Data services such as the Internet, IMS, Operator services, Application specific services etc.
- Connections are typically through different i.e. Multiple PDU sessions
  - Each going to one different DNNs,
  - via 3GPP or Non3GPP or both,
  - connecting through same or different set of 5GC NFs

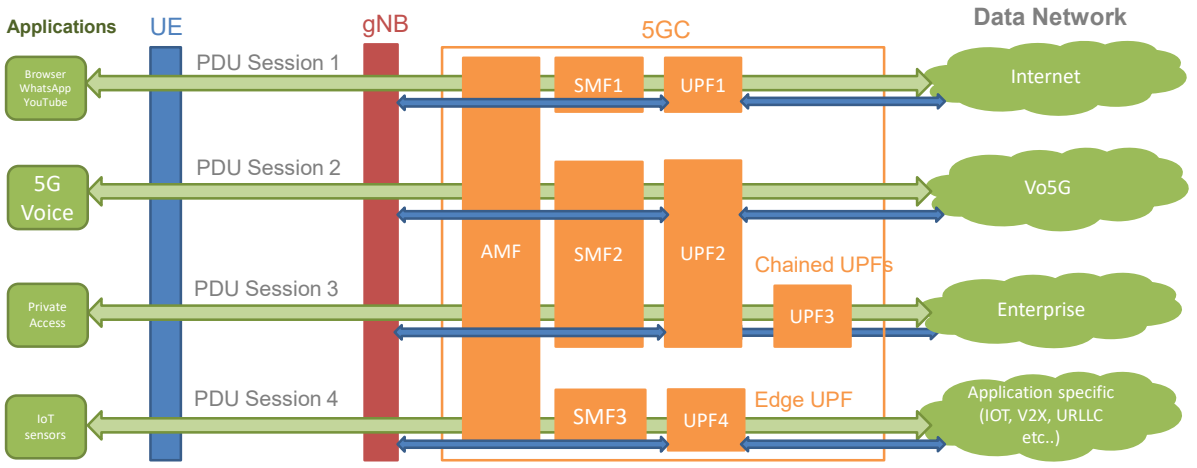
Data network connectivity

Connections to different data networks for different services
Through multiple (simultaneous) PDU sessions



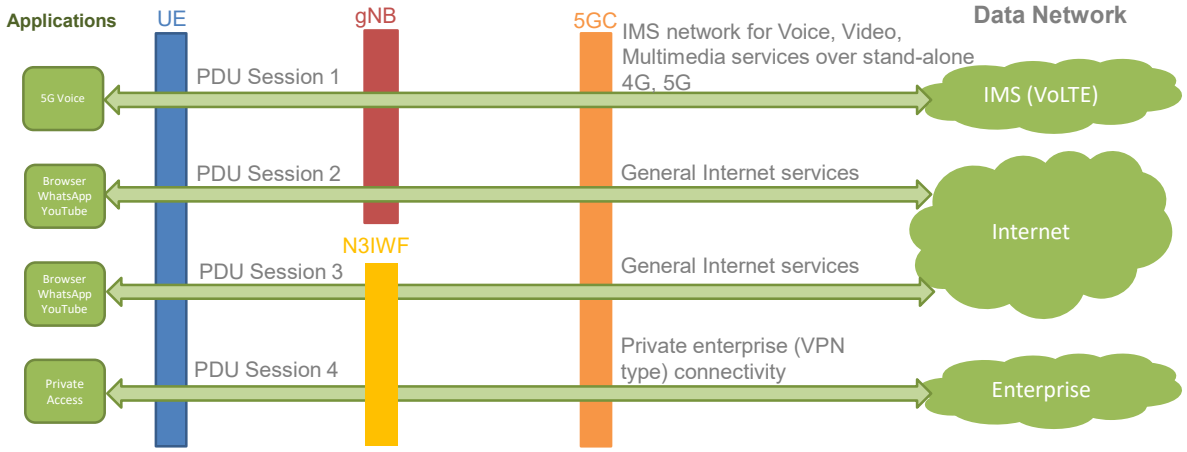
5GC Components

- ✓ Each UE will have a single AMF
- ✓ Each PDU session can be through different SMFs (registered per PDU session in the UDM)
- ✓ Each PDU session can go through one or more UPFs



Multiple Access

- Connectivity over both 3GPP and Non-3GPP Access
- Can be for different PDU Sessions or the same PDU session
- Can be simultaneous
- UE move across AN per PDU session basis



SM Services

- Session Management (SM): Enables exchange of data/PDUs between the UE and DNN by setting up and maintaining the required signaling and data paths
- PDU Session Establishment: always UE initiated
  - Sets up the signaling path, negotiates the session parameters
  - IP address assignment is done as part of this
  - Can be network triggered
- Modification: initiated UE or 5GC
  - To add/modify/delete QoS
  - To change PS data off or always-on options
- Release: upon UE or 5GC request
  - Teardown the signaling path when it is no longer needed

PDU Session Attributes

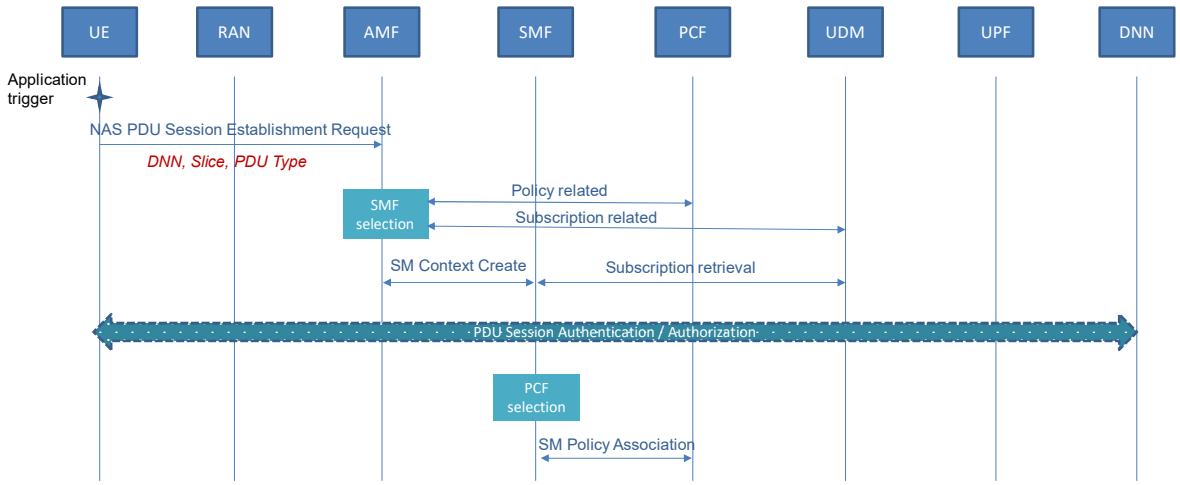
Key parameters of a PDU session

- Sent by UE
- Network compares against subscription information and assigns accordingly

- PDU Session ID
  - Generated by UE, unique per session
- PDU Session Type
  - One of: IPv4, IPv4, IPv4v6, Ethernet, Unstructured
- S-NSSAI
  - Desired Slice
- DNN
  - Target data network

- SSC mode: Service and Session Continuity mode
  - SSC 1: (seamless) Connectivity is preserved (for IP sessions, IP address is preserved)
  - SSC 2: (break) Network may release PDU sessions
  - SSC 3: (make-before-break) Connection through new PDU Session Anchor is made before the existing connection is torn down (IP address not preserved). Network preserves connectivity and leave it to UE to adapt to changes
- User Plane Security Enforcement Information
  - Integrity and Ciphering rules: Required/Preferred/Not needed.

PDU Session Setup



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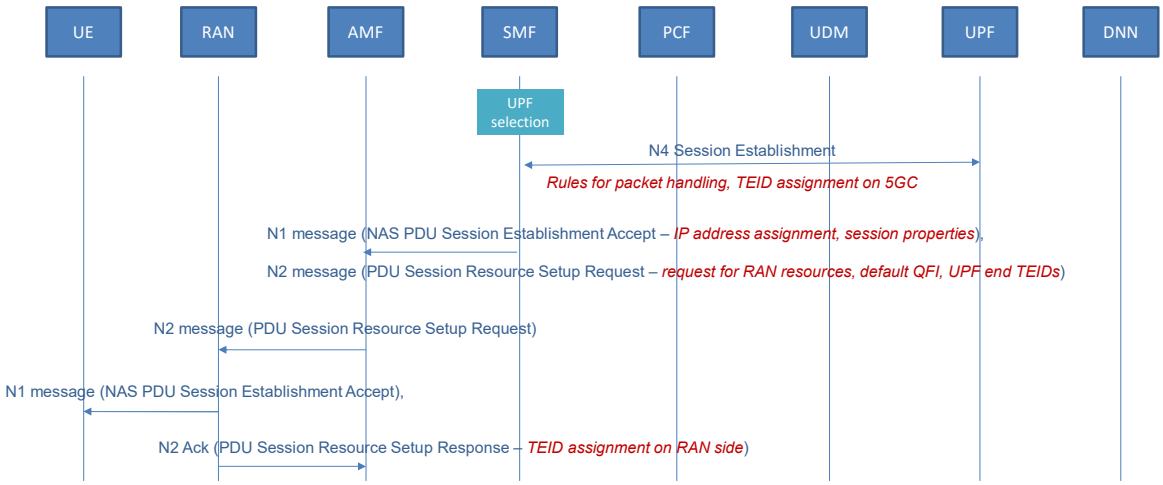
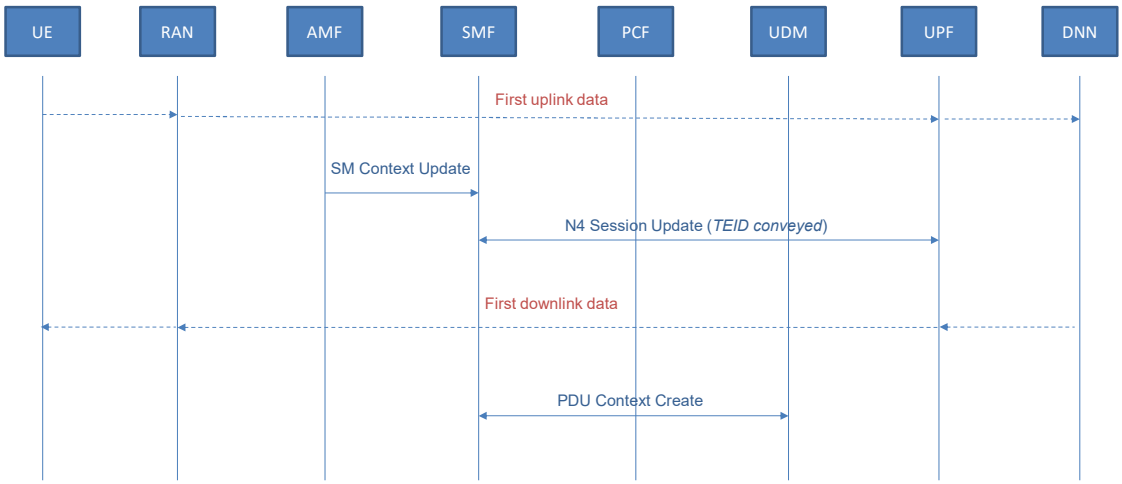


PDU Session Setup



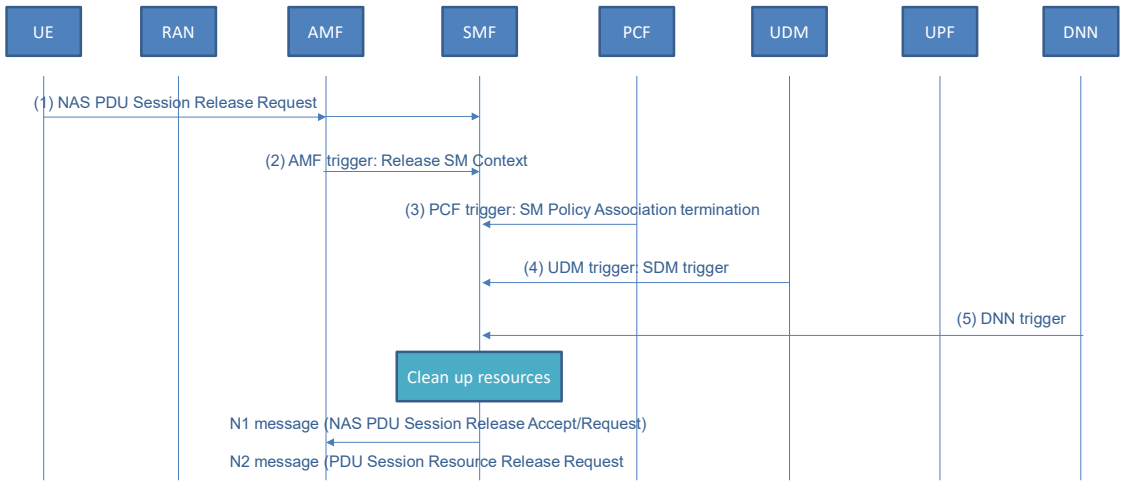
PDU Session Release - Why

- When application no more needs the services of 5GS
- To release
  - all resources associated with the PDU Session
  - IP address/Prefixes allocated for an IP-based PDU Session
  - All UPF & RAN resource used by the PDU Session
- Both UE and network (SMF) will notify all dependent entities

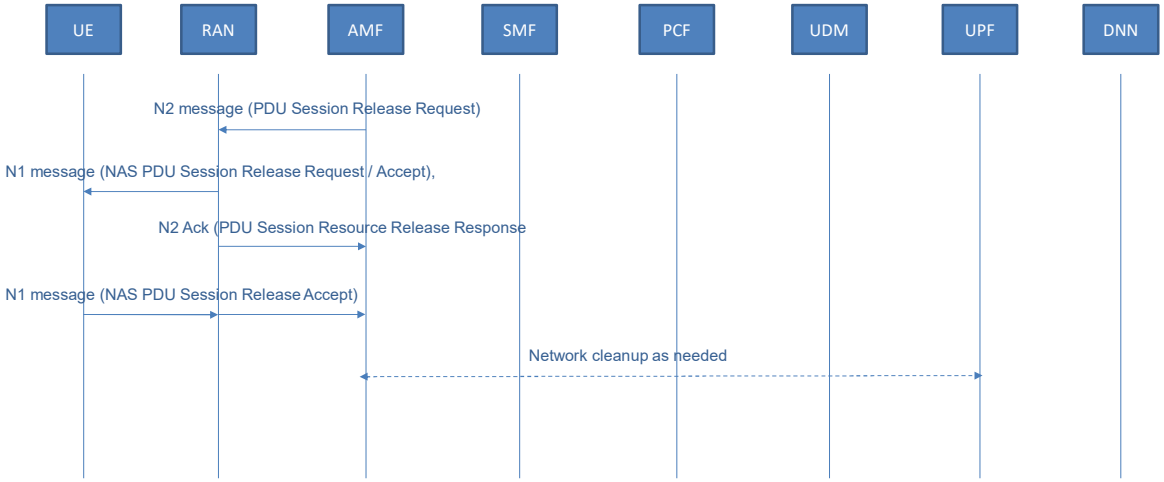
PDU Session Release initiation

- UE triggered
  - When all applications using that IP address have exited and connection is no longer needed
- Network triggered
  - AMF triggered in case of mismatch of PDU Session status between UE and AMF
  - PCF may invoke an SM Policy Association Termination procedure to request the release of the PDU Session
  - (R)AN may indicate to the SMF that the PDU Session related resource is released, e.g. when all the QoS Flow(s) of the PDU Session are released
  - On request from the UDM (subscription change) or from the CHF
  - On request from the DNN (cancelling the UE authorization to access to the DN)
  - When UE moves out of LADN service area, AMF indicates to SMF
  - Based on locally configured policy (e.g. the release procedure may be related with the UPF re-allocation for SSC mode 2 / mode 3)

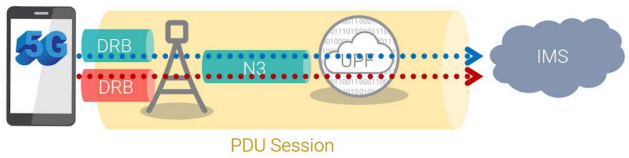
PDU Session Release

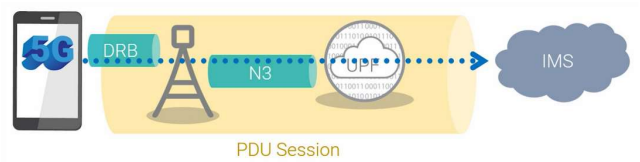


PDU Session Release



PDU Session Modification





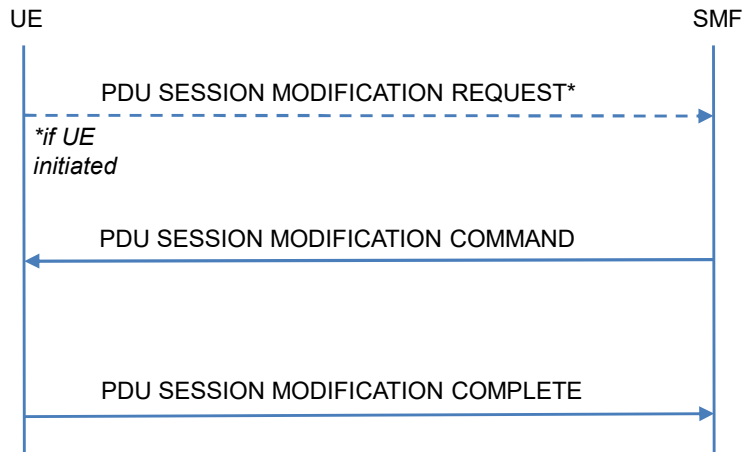
PDU Session Modification

- UE initiated
  - To request specific QoS handling and segregation of service data flows (Add/modify/delete QoS flows)
  - To indicate a change of 3GPP PS data off UE status for a PDU session
  - To revoke the previously indicated support for reflective QoS
- Network initiated
  - PCF triggered (policy change, AF triggered, management triggered)
  - UDM initiated (subscription change)
  - SMF initiated (local policy, triggered by RAN, UPF related changes)
  - AN triggered

Note:

- UE & AN can only request, decided and controlled by network

PDU Session Modification: NAS Call Flow



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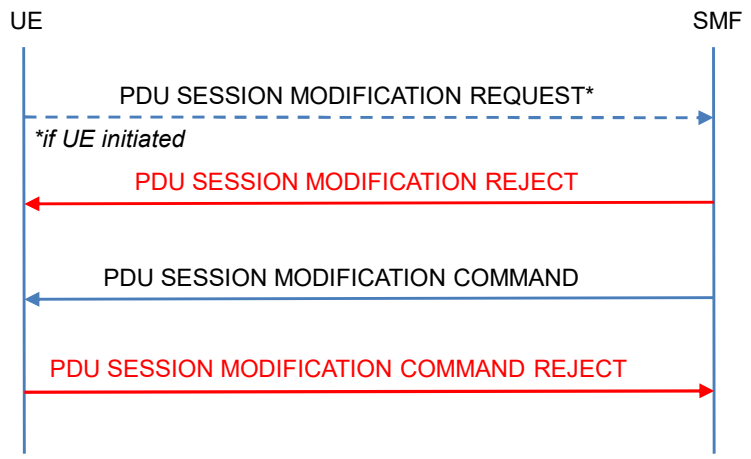
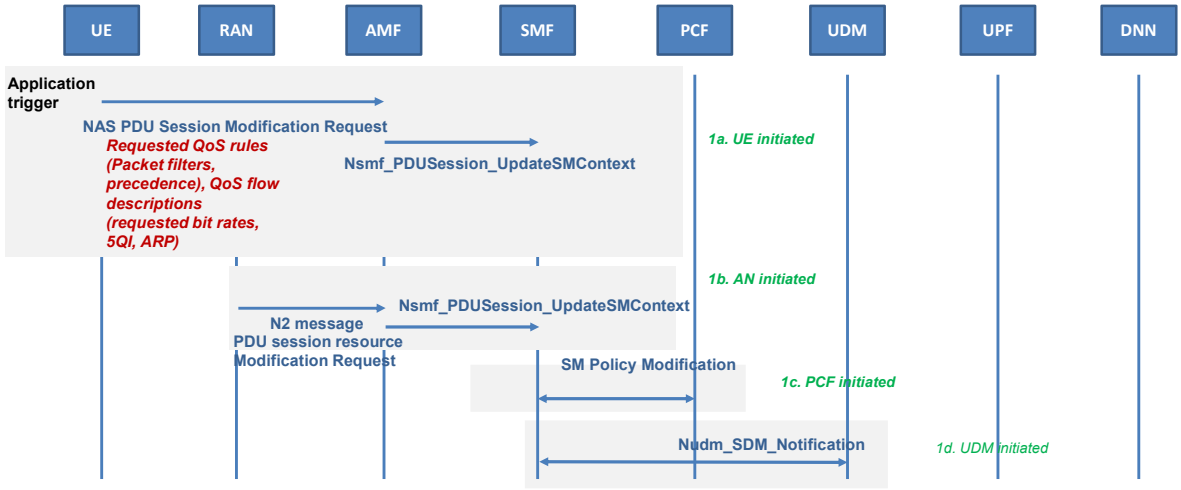
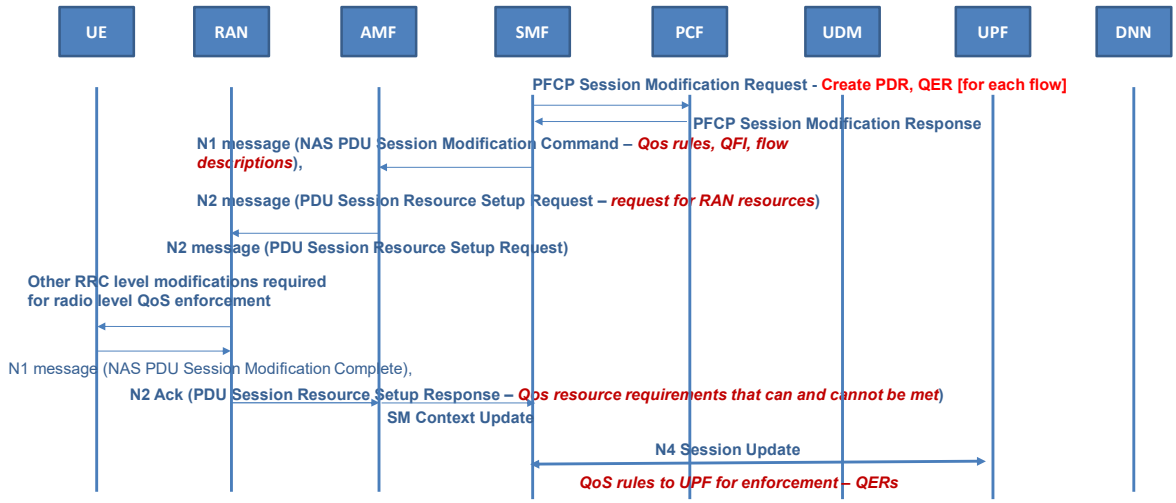


PDU Session Modification - Triggers



PDU Session Modification



THANK YOU